

Write each pair of parametric equations in rectangular form.

5) $x = -\frac{t^2}{3}, y = t$

6) $x = t, y = \frac{t^2}{6} + \frac{2t}{3} - \frac{1}{3}$

7) $x = -2t - 3, y = 2t^2 + 2t - \frac{5}{2}$

8) $x = 2\sin t, y = 4\cos t$

9) $x = \sec t, y = 4\tan t$

10) $x = 4\cos t - 1, y = 3\sin t + 1$

Use the parameter to write each rectangular equation as a pair of parametric equations.

11) $x = \frac{y^2}{6}, t = y$

12) $y = -\frac{x^2}{4} + x + 1, t = -\frac{x}{3} + \frac{1}{3}$

Critical thinking questions:

13) Write a set of parametric equations that represent $y = x^2 - 4x$. Then write a second set of parametric equations that represent the same function, but with a slower speed

14) Write a set of parametric equations that represent $y = x^2 - 1$. Then write a second set of parametric equations that represent the same function, but with a faster speed and an opposite orientation.